Bacterial Vaginosis

Karen Pendergrass, Divine Aleru

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Abstract. Bacterial vaginosis is caused by an imbalance in the vaginal microbiota, where the dominant Lactobacillus species are significantly reduced.

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Overview

Bacterial vaginosis (BV) is caused by an imbalance in the vaginal microbiota, where the typically dominant Lactobacillus species are significantly reduced, leading to an overgrowth of anaerobic and facultative bacteria such as *Gardnerella vaginalis*, *Atopobium vaginae*, and *Prevotella spp*. [1] This results in symptoms like discharge, odor, and discomfort, and increases susceptibility to other infections, including sexually transmitted infections like HIV/AIDS. [2][3] The intricate relationships among these bacteria vary among individuals, complicating the understanding and treatment of BV. Advances in molecular techniques have improved the identification and understanding of these microbial communities, which is essential for developing effective treatments and preventative measures for BV and its complications.

Diagnosis

BV can be diagnosed using Amsel criteria and Nugent score through vaginal swabs. Amsel criteria require three of the following four characteristics: homogeneous, milk-like vaginal fluid, increased vaginal pH (greater than 4.5), fishy odor, and clue cells (vaginal epithelial cells covered with bacteria). [4][x] Nugent score involves analyzing morphotypes of microorganisms like *Lactobacillus* and *Gardnerella* in Gram-stained vaginal discharge, with scores of 0–3, 4–6, and 7–10 indicating normal, intermediate, and BV, respectively. [5] However, 27% of asymptomatic women show a diverse microbial community dominated by anaerobic bacteria. [6] Combining these methods with clinical and microbiological findings provides a more accurate diagnosis.

Causal Theories

BV remains a challenging condition due to its multifactorial etiology. Several theories exist about what triggers BV, each with its own set of implications, strengths, and criticisms.

Theory	Description	Criticisms and Limitations
Polymicrobial Hypothesis	BV is caused by an overgrowth of anaerobic bacteria due to a shift away from Lactobacillus dominance.[7]	Fails to identify a single pathogen and does not explain why different microbial compositions are found in different individuals.[8]
Gardnerella vaginalis as Causative Agent	*Gardnerella vaginalis* is often considered the primary cause of BV due to its frequent presence in affected women.[9]	*Gardnerella* can also be found in women without BV, suggesting that it may not be the sole causative agent as it is part of the normal vaginal microbiome, even in healthy women.[10]
Sexual Transmission Hypothesis	BV may be sexually transmitted, with evidence of microbial exchange between partners.[11][12]	Past partner treatment trials have had mixed results, reducing confidence in the idea that reinfection drives recurrence.[13]
Host Factors and Immune Response	Immune responses and genetic predisposition may influence susceptibility to BV.[14][15]	The role of immune factors in BV pathogenesis is still speculative, and there is no clear explanation for recurrence despite similar immune profiles.

Given the limitations of current theories, a new theory that combines insights from microbial interactions, host immunity, and sexual transmission could provide a more comprehensive understanding of BV.[16]

Associated Conditions

BV has been closely associated with various reproductive health complications, including infertility, pelvic inflammatory disease (PID), and endometritis.[17] BV is marked by a microbial imbalance in the vaginal flora, where the predominance of *Lactobacillus* species is replaced by an overgrowth of anaerobic bacteria, such as *Gardnerella vaginalis* and *Atopobium vaginae*. This dysbiosis is thought to increase the risk of PID, which in turn is linked to infertility, particularly tubal infertility.[18] Women with BV are more likely to experience chronic inflammation in the reproductive tract, which may impair fertility through mechanisms such as immune system activation and the production of bacterial toxins. Studies have shown that BV-related bacteria can ascend to the upper genital tract, causing infections like endometritis and contributing to conditions that affect the fallopian tubes and uterus, leading to infertility. BV is also known to cause adverse pregnancy outcomes, and increased susceptibility to HIV. [19]

Primer

To fully understand Bacterial Vaginosis (BV), it is crucial to recognize its complex relationship with the vaginal microbiome, where a loss of *Lactobacillus* species and an overgrowth of anaerobic bacteria disrupt the normal balance. This dysbiosis leads to increased vaginal pH and inflammation, contributing to symptoms such as discharge and malodor. BV is not only associated with reproductive issues like infertility and pelvic inflammatory disease (PID), but it also raises the risk of acquiring sexually transmitted infections, including HIV. The persistence of BV is often linked to biofilm formation, where bacteria protect themselves from antibiotics and the host immune response, making recurrence common. Understanding these microbial shifts, the role of biofilms, and their interactions with immune and hormonal factors is

essential for developing more effective treatments and diagnostic tools to manage BV and prevent its recurrence.[20][21][22]

Metallomic Signature

Heavy metal exposure, particularly to lead (Pb) and cadmium (Cd), has been shown to significantly increase the risk of developing Bacterial Vaginosis (BV). Studies have shown a notable association between elevated serum lead and cadmium levels and a higher likelihood of BV. Specifically, serum lead levels in the highest range were associated with an increased risk of BV, while higher cadmium levels were linked to a rise in susceptibility. These metals, known as endocrine-disrupting compounds (EDCs), disrupt immune function and can lead to hormonal imbalances, which in turn affect the vaginal microbiome. Elevated lead and cadmium levels may foster an environment in which anaerobic bacteria thrive, thereby exacerbating the dysbiosis typical of BV. These findings highlight the potential impact of reducing heavy metal exposure to manage and prevent BV, highlighting the need for further research into how these metals influence the vaginal microbiome and immune system.[23]

Nutritional Immunity

Iron is a key player in BV, as increased iron availability, particularly from menstruation or dietary sources, can promote the growth of BV-associated anaerobic bacteria. These bacteria thrive in iron-rich environments, which in turn exacerbates the dysbiosis in the vagina. Conversely, an iron deficiency can weaken the immune response, making it more difficult for the body to control infections, increasing susceptibility to BV.[24][25] The complex interplay of this nutritional factor affects how the body regulates the vaginal microbiome, immune system, and local inflammation. The relationship between nutritional factors and BV highlights the importance of maintaining a balanced diet to support immune health and prevent the onset or recurrence of BV.

Interventions

Microbiome-targeted interventions (MBTIs) focus on restoring the balance of the vaginal microbiota to treat bacterial vaginosis (BV). These interventions aim to reduce the overgrowth of obligate anaerobic microbes, such as *Gardnerella*, *Prevotella*, and *Atopobium*, which are associated with BV. At the same time, MBTIs seek to increase the presence of beneficial *Lactobacillus* species, which are crucial for maintaining a healthy vaginal environment. By targeting the microbial imbalance directly, MBTIs offer a more precise approach to treating and preventing BV compared to traditional antibiotic treatments.

▶ **What microbiome-targeted interventions may improve symptoms of BV?**

Intervention	Classification	MBTI Status
Metronidazole	Pharmaceutical	Validated
Secnidazole	Pharmaceutical	Validated
Tinidazole	Pharmaceutical	Validated
Clindamycin	Pharmaceutical	Validated
Boric Acid	Alternative Therapy (Non-pharmaceutical)	Validated
Probiotics	Supplement	Validated
Essential Oils	Natural Product	Promising Candidate
Vitamin C (Intravaginal tablets)	Supplement	Validated
Cannabidiol	Natural Product	Promising Candidate
Maltose gel	Prebiotic	Validated
Lactoferrin	Supplement	Validated
Metformin	Drug Repurposing	Promising Candidate
Vaginal Microbiota Transplantation (VMT)	Microbiota-based therapy	Promising Candidate

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